

Family Transitions in Europe: From Leaving Parental Home to First-Time Parenthood Through a Macro-Policy Lens

Michaela Bunakova¹

¹*Department of Sociology, McGill University, 885 Sherbrooke Street West, Montreal, Quebec, Canada, H3A*

Corresponding Author: Michaela Bunakova, MSc, IRIS Laboratory, Peterson Hall, McGill University, 3460 Mc Tavish St, Montreal, Quebec H3A 0E6. Email: michaela.bunakova@mail.mcgill.ca

Contributors: MB was responsible for the concept, design, and main analysis of this study, as well as the written draft of the final manuscript.

Funding: This study was supported by the Canada Research Chair in Policies and Health Inequalities (PI: AQV).

Conflicts of interest: The author declares that there are no conflicts of interest relevant to the content of this article.

Availability of data and material: The ESS Round 9 dataset is publicly available for download from the European Social Survey. Data analysis and code from STATA, version 15, are available upon request.

Ethics approval: TCPS 2 Statement, Article 2.2: research relying exclusively on publicly available, encrypted and anonymized data such as the ESS Round 9 dataset is exempt from formal TCPS2 ethics review.

Consent to participate and publish: Not applicable.

Abstract

Objectives: This study attempts to deepen our understanding of the risk of becoming a first-time parent following an exit from parental home, with a particular emphasis on the interconnection between individual-level demographic factors, expectations about family formation and macropolitical factors in Europe.

Methods: We used survey data from 30 European countries from 2018 Round 9: Timing of life of the European Social Survey (ESS9). Using event history analysis and comparative social policy analysis methodology, we estimated the risk of becoming a first-time parent in Europe following the separation from living with parents. We estimated a set of six weighted proportional hazard regressions for survey data and presented key macro-level variables on Kaplan-Meier Survival Curves.

Results: We found that individual-level sociodemographic characteristics of respondents consistently affect the risk of becoming a first-time parent, whereby the risk is lower for males, individuals who have never been in a formal relationship, with more years of education, higher mother's education and higher occupational class, compared to females, those who declared a formal relationship in the past, with lower personal and mother's educational attainment and lower occupational class, respectively. Additionally, we found that macro-level indicators of family-friendly policies matter strongly for the respondents' decision to become parents and more generous benefits are associated with a higher probability of having a first child following a move away from parental home.

Conclusions: Our findings highlight an important unrecognized potential of the UNICEF's family policy recommendations to affect not only the wellbeing of children, but also the decision to become parents at a given time in the first place. Future research needs to examine fertility decisions and family transitions through the lens of access to affordable family-friendly policies.

Keywords: *First-time parenthood, housing independence, welfare regime, maternity leave duration, guaranteed childcare, event history analysis*

1 Background and Rationale

Over the last decades, we have witnessed decreasing fertility rates in Europe, as well as a later average age of first-time parenthood (Frejka and Sardon 2006). At the same time, individuals in more recent cohorts of adults tend to leave their parental homes later in life, marry later and are less likely to afford to purchase a home, compared to earlier post-war birth cohorts (Buck and Scott 1993, Mulder and Hooimeijer 2002, Elliott et al. 2012, Dewilde and Lancee 2013). This has important implications for the demographic composition of populations, the projected workforce, policy reforms and planning. It is crucial to understand the trends, potential mechanisms and core underlying factors affecting the likelihood of becoming a first-time parent after establishing independence and moving out from the parental home. This is because the processes of family planning after establishing independence

from one's parents may be significantly different to those of individuals who are still living with their own parents. Individuals who moved out to live separately from their parents may face additional burdens related to their macropolitical environment, such as financial independence, housing affordability, demands from employment or inability to plan for sufficient childcare, that may affect their ability to form a family and afford to have a child.

This paper aims to explore both individual-level and macropolitical factors associated with the transition process from leaving the parental nest to becoming a first-time parent in Europe. While studies that document patterns of leaving home or fertility rates using similar event history techniques exist, such studies are both outdated and either address factors associated with leaving parents or becoming parents in one analysis, but not the transitions between the two (Buck and Scott 1993; Heaton and Call 1995; Mulder, Clark, and Wagner 2002; Mulder and Hooimeijer 2002; Studer, Liefbroer, and Mooyaart 2018; Van Hook and Altman 2013). This paper attempts to

deepen our understanding of the risk of becoming a first-time parent following an exit from parental home, with a particular emphasis on the interconnection between individual-level demographic factors, expectations about family formation and macropolitical factors. This study could help us map how family formation differs across time-points, genders, or regions with different family policy implementation strategies. These policies include parental leave duration, availability, and affordability of early childcare services, identified as crucial according to UNICEF's 'Early Moments Matter' OECD policy campaign (Chzhen, Gromada, and Rees 2019). While multiple studies exist that estimated the effect of these policies on the development and long-term well-being of children, to the author's knowledge, no study estimating the impact of these policies on the decision to become a first-time parent currently exists (Chzhen et al. 2019).

Research Question

What factors are associated with the likelihood of becoming a first-time parent after gaining independence and moving out of a parental home?

Hypothesis

We hypothesize that the risk of becoming a parent of a first child following one's separation from living with their parents is associated with varying personal expectations about the ideal age to both become parent and move out of one's parental home. We also hypothesize that the macropolitical environment in which individuals live, including the availability and generosity of parental leave or childcare services, matters for family formation trajectories and that those who live in countries with more generous family policies will choose to have children earlier than otherwise-demographically-similar individuals living in countries with more stringent family policies.

2 Methodology

Data

We use the 2018 Round 9: Timing of life of the European Social Survey (ESS9), a bi-annual, cross-sectional international survey of 30 European countries. The ESS9 involves rigorous translation protocols for maximum cross-country comparability and is conducted using hour-long face-to-face CAPI interviews (European Social Survey 2018b). The ESS9 dataset is publicly available, encrypted, and anonymized. The ESS9 frame for survey is based on strict random probability sampling, representative of all persons aged 15 and over in Europe with an effective minimum sample size of at least 800 individuals (European Social Survey 2018a). The survey uses a balanced response rate strategy across countries, with 70% representing the minimum target response rate in each country surveyed, and to adjust for unequal representation of subgroups, post-stratification and population-level weights are provided and used in this analysis.

Target population and definition of analytic sample

The target population for this analysis are all European individuals aged 15 and older at risk of becoming first-time parents following a reported separation from living with parents. By survey design, ESS9 included only individuals aged 15 and older, so no age restrictions on the dataset needed to be made. Based on the survey question 'In what year, if ever, did you first leave your parent(s) for 2 months or more to start living separately from them?', we excluded individuals who stated that they are still in parental home, who never lived with parents, or those for whom a response this variable is missing from our analytic sample. Only complete cases with respect to all main variables were included in the analysis and missing values were uniformly recoded and excluded using list-wise deletion. This means that from the total of 49,519 ESS9 respondents, we excluded 38,490 individuals, or 78% of the total sample, yielding an analytic sample of 11,029 unique respondents.

Analytic method

To estimate the likelihood of having a first child after leaving a parental home to start a separate, independent life, we used event history analysis methodology, and particularly a set of six weighted proportional hazard regressions for survey data. The six regression models start from the simple model, which includes only individual-level sociodemographic variables, and then incorporate different expectation-based and macro-level variables one-by-one, until the full model, which includes all variables of interest and relevant interaction terms. Additionally, we portray our data graphically, using four Kaplan Meier survival curves for all macro-level variables of interest.

We have defined and coded our event of interest as 1 if event happened (i.e., a respondent reported a first child's year of birth) and 0 if the respondent never had any children. Our event history duration was computed as follows:

- for individuals who reported a year during which their first child was born as: *year a respondent became parent – year of first move from parent's house*
- for individuals who reported that they never became parents: *2018 – year of first move from parent's house*

The follow-up of respondents was ended after 25 years since the year individuals stopped living with their parents, because this follow-up duration approximates the at-risk period for becoming a first-time parent. For example, if an individual left to live separately at the age of 15, this person will be either followed for 25 years, until the age of 40, or until the event of interest, the birth of the first child is experienced, at which point she will be censored out of the analysis.

Variables

The main outcome of interest, based on our duration and event variable, is the risk of becoming a first-time parent after leaving one's parental home. The independent variables in this analysis consisted of 8 individual-level sociodemographic vari-

ables, two personal expectation questions and four constructed macro-level policy variables.

Sociodemographic covariates. Our sociodemographic independent variables included: a dummy variable for marital status (1 if a respondent was ever married or in a civil partnership and 0 if an individual never had a formally recognized relationship at the time of survey); a continuous variable for age at which a respondent left their parents' house; a categorical variable for a respondent's birth cohort (born between 1929–49, 1950–69, 1970–89, 1990 and over), computed from the respondent's age at the time of survey; a continuous variable for self-reported household income at the time of survey; a dummy variable for self-reported gender (male or female); a continuous variable for the number of years of a respondent's full-time education; a continuous variable for the respondent's mother's highest level of education; a categorical variable for occupational status, computed from the ISCO08 occupation classification (1 = managerial work, 2 = professional work, 3 = technical, clerical, and associate professional work, 4 = skilled agricultural and trade work).

These sociodemographic covariates were chosen with reference to established research evidence on determinants of both family formation trajectories and leaving parental home (Buck and Scott 1993; Heaton and Call 1995; Mulder et al. 2002; Mulder and Hooimeijer 2002; Studer et al. 2018; Van Hook and Altman 2013).

Expectations about timing of life. We have included two individual-level variables, aim of which is to approximate respondents' expectations about ideal age to leave parental home and become parents. These variables include: a continuous variable based on survey question 'In your opinion, what is the ideal age for a girl/boy or woman/man to become a mother/father?'; a continuous variable based on survey question 'After what age would you say a woman/man is generally too old to still be living with her/his parents?'. By including these variables, we hope to control for different cultural and personal expectations about timing of life events, which may vary between regions and may otherwise confound the estimates of our macro-level variables.

Macro-level welfare regime and policy covariates. Lastly, we have computed and coded four separate macro-level explanatory variables to classify respondents' countries of residence based on their family-friendly policies and/or welfare regime categorization. These variables include: a categorical variable for welfare regime type, based on an adapted regime typology, including elements of Esping-Andersen's original three world typology (Esping-Andersen 1990), Ferrera's additions (Ferrera 1996) and Eikemo's modifications (Eikemo et al. 2008), such that 4 distinct regimes were identified, the Scandinavian, Continental/Conservative, Mediterranean, Central/Eastern European and Anglo-Saxon (see Appendix A for detailed classification of each country included and their position on the full state-family-market spectrum of service provision); a continuous variable for national mandated maternity leave duration in 2018, as measured in weeks of leave from employment and entitlement to either full or partial compensation, coded and sourced from the OECD Family Database PF2.1 on Parental

Table 1. Summary statistics for main individual-level variables of interest

	Mean	SD	Range
Marital Status	1.502	.500	1
Age left home	20.412	4.491	65
Household income	5.455	2.768	9
Birth year	1966.4	16.927	75
Gender	1.547	.498	1
Years of education	13.456	4.186	60
Mother's education	2.743	1.747	6
Occupational class	2.586	1.213	3
Ideal age, 1st parenthood	27.281	3.652	50
Age too old to live w/ parents	28.562	6.625	105

Leave Systems (OECD 2020); a dummy variable for whether a respondent's country of residence offered a guaranteed national early childcare for children under 3, at the time of survey, coded and sourced from the European Commission's Eurydice data report (Motiejūnaitė-Schulmeister, Balcon, and De Coster 2019); a dummy variable for whether a respondent's country of residence offered a fully free national early childcare service for children under the age of 3 at the time of survey, coded and sourced from the European Commission's Eurydice data report (Motiejūnaitė-Schulmeister et al. 2019).

Welfare regime typology literature suggests that individual regimes are based on different reliance on the state-family-market distribution of services and responsibilities, whereby some regimes, such as the Conservative Regime, rely more heavily on families for provision of care, than other regimes, such as the Scandinavian, where care is more outsourced to the responsibility of the state (Aspalter 2020). Using welfare regimes is particularly useful because it allows us to categorise countries according to 'ideal types' of welfare institution design, which allows us to take advantage of only 5 categories of regimes with similar welfare provision attributes instead of 30 individual countries, which makes our statistical study easier to conduct and interpret. This offers us valuable comparative insights into the effect particular designs of welfare provision have on the uptake of caring responsibilities in families and health inequalities in our societies.

The three direct measures of family-friendly policies are exploring whether common national family policies, aim of which is to improve the wellbeing of the parents and children alike as well as help reconcile motherhood with employment opportunities, affect the decision to become a first-time parent in the same way across countries. Empirical evidence from comparative literature suggests that policies that intent to reduce the opportunity cost of childbearing, affect fertility both directly and indirectly (Esping-Andersen 2007). For example, if maternity entitlements are too short, and high-quality early childcare is neither available nor affordable, women with low education may abandon employment, while highly educated women may limit or forgo having children (Esping-Andersen 2007).

3 Results

Table 1 above shows the summary statistics, including mean, standard deviation and range for all individual-level variables

of interest. We see that about 50% of our sample have been married or in a formal partnership in the past, since the dummy is coded as 1=ever married/cohabiting and 2=never married/cohabiting. The average age at which respondents left their parental home was 20.4 years. The average income in the 5th income decile on a scale of 10 deciles. There are slightly more females than males in our sample, with an average of 13.5 years of full-time education. On average, the highest completed education level of respondents' mothers was lower tier upper secondary school. The average occupational class was professional and associate professional work. The mean ideal age for first time parenthood is 27.3 years and the age at which respondents consider one to be too old to still live with one's parents is on average 28.6. It is important to note that the ranges of values for some of the variables are very large. For example, the range for the last variable is 105, indicating that some respondents believe that there is no age at which it is too old to live with one's parents. Additionally, Appendix B and Appendix C present the summary statistics stratified by welfare regime types and birth cohorts, to check whether there are significant differences in the main individual-level variables across time and regional clusters. We found, as expected, that there is a significant difference between birth cohorts with respect to marital status, whereby with each more recent birth cohort, the mean proportion of respondents who reported having been married or cohabiting significantly decreases. Birth cohorts do not significantly differ with respect to the gender distribution, mean age at leaving parental home, ideal age to become a first-time parent or age at which one is too old to still be living with one's parents. When it comes to differences by welfare regimes, there is a slightly lower proportion of married individuals and significantly lower age at which respondents believe one is too old to still live with parents in the Scandinavian regime countries compared to the other regimes. Central and Eastern European regime countries have significantly higher mean maternity leave duration compared to the other regimes. No significant difference between welfare regimes was found with respect to mean age at which respondents left parental home, years of education, average respondent age, gender, and ideal age to become first-time parents.

Discussion of Findings

In the table below (Table 2), we present 6 proportional hazards regression models. The first model includes only individual-level and shows that the risk of becoming a first-time parent following separation from parental home is 72% lower for respondents who have never been married or cohabiting, compared to those with past reported relationship, keep all else constant. This estimate remains very similar in terms of magnitude and statistical significance across all models.

The effect of age at which respondents left parental home, while important as an adjustment variable, does not appear substantially important across all models. The only statistically significant difference among birth cohorts appears to be between those born in 1970 to 1989 and after 1989, across all 6 models. Those born between 1970 and 1989 were at 27% higher risk of becoming first time parents after leaving

parental home than those born after 1989. Appendix D includes an additional Kaplan-Meier Survival curve for the surviving, before becoming a first-time parent across birth cohorts and showed a clear gradient trend of higher survival for more recent birth cohort. However, as our model suggests due to the lack of significance of the birth cohort covariate, this difference was likely explained by factors other than respondents' birth cohorts. Females were significantly more at risk of becoming parents after leaving parental home than males in all models. The effect was similar in magnitude across models 1 to 5, but in Model 6, the effect has to be interpreted taking included interactions in considerations. There appears to be a gradient in the risk of becoming first-time parents across occupational status categories, whereby each lower occupational category has a higher risk of becoming first-time parent compared to the managerial occupational position.

In Model 2, we added two expectation-based individual-level variables, one for the ideal age to become a parent and one for age a respondent considered too old to live with parents. The latter is neither substantively, nor statistically significant, while the former appears to slightly reduce the risk of becoming a first-time parent, such that for each extra year of the ideal age to become a parent, the risk of becoming a first-time parent is lower by 4%. The magnitude of this effect remains the same in the full model (Model 6), but the statistical significance is reduced from $p < 0.001$ to $p < 0.05$. In Model 6, we also included an interaction term for the ideal age and gender, and we found that compared to males, each additional increase in the ideal age to become parents is associated with a reduced risk of becoming parents for females, keeping all else constant.

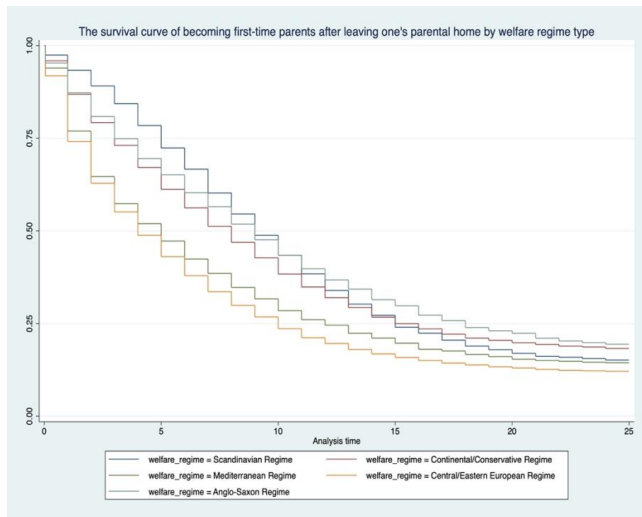
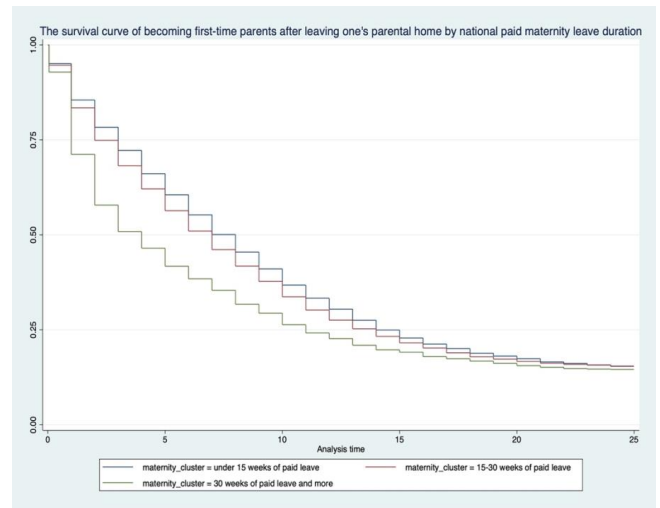
Model 3 included our categorical variable for welfare regime type. By including the computed macro-level variable, the effect size and significance of all individual-level covariates previously included, remained equally robust. Our model suggests that compared to those who live in Central or Eastern European welfare regime, individuals who live in all other regimes are at significantly lower risk of becoming a first-time parent than those in the Central/Eastern European regime.

Figure 1 shows these findings graphically, pointing to higher survival before becoming first time parents for respondents living in the Scandinavian Regime at all durations of the follow-up, ending with the Central/Eastern European Regime whereby respondents have the highest risk of becoming first time parents among the other regimes. In Model 4, we substitute our welfare regime variable with national-level maternity leave duration. Much like with the case of the welfare regime variable, the estimate sizes for the individual-level sociodemographic predictors remained the same, but this time, the model suggests that each additional national-level week of maternity leave is associated with a 1% increase in risk of becoming first time parent. From Figure 1, we see that when clustered into three maternity leave duration categories, having under 15 or 15-30 weeks of paid leave makes little difference to the survival, while having over 30 weeks available significantly affects the risk of becoming a first-time parent.

Model 5, on the other hand, substitutes the maternity leave

Table 2. Proportional Hazards regression results (exponentiated coefficients; t statistics in parentheses; * p<0.05, ** p<0.01, * p<0.001)**

	(1) Individual	(2) 1+Expectations	(3) 1+Regimes	(4) 1+Maternity	(5) 1+Childcare	(6) Full Model
Is/was ever married	1	1	1	1	1	1
Never married/cohabiting	0.281*** (-19.77)	0.282*** (-19.17)	0.284*** (-19.78)	0.281*** (-19.90)	0.280*** (-19.92)	0.282*** (-19.74)
Age left parental home	1.011* (2.05)	1.013* (2.32)	1.010 (1.83)	1.008 (1.60)	1.011* (2.17)	1.021*** (2.84)
Birth cohort 1929-49	1.046 (0.33)	1.007 (0.05)	1.038 (0.27)	1.050 (0.35)	1.043 (0.31)	0.994 (-0.04)
Birth cohort 1950-69	1.000 (-0.00)	0.995 (-0.04)	1.001 (0.01)	1.003 (0.03)	0.996 (-0.03)	0.991 (-0.07)
Birth cohort 1970-89	1.267* (2.08)	1.276* (2.16)	1.271* (2.11)	1.268* (2.08)	1.269* (2.09)	1.276* (2.18)
Birth cohort 1990-09	1	1	1	1	1	1
Household Income	1.029* (2.55)	1.035** (2.95)	1.030** (2.67)	1.030** (2.64)	1.030** (2.58)	1.035*** (3.09)
Male	1	1	1	1	1	1
Female	1.718*** (9.73)	1.808*** (10.47)	1.712*** (9.76)	1.714*** (9.72)	1.720*** (9.79)	5.027*** (3.20)
Years of education	0.965*** (-5.25)	0.969*** (-4.58)	0.966*** (-5.11)	0.966*** (-5.16)	0.965*** (-5.24)	0.971*** (-4.45)
Mother's Education	0.961* (-2.41)	0.962* (-2.34)	0.953** (-2.84)	0.962* (-2.35)	0.959* (-2.53)	0.955** (-2.76)
Occupational class 1	1	1	1	1	1	1
Occupational class 2	0.949 (-0.64)	0.964 (-0.45)	0.975 (-0.30)	0.950 (-0.63)	0.952 (-0.60)	0.978 (-0.28)
Occupational class 3	1.205** (2.65)	1.202** (2.59)	1.215** (2.80)	1.211** (2.74)	1.207** (2.69)	1.206** (2.70)
Occupational class 4	1.356*** (3.87)	1.367*** (3.94)	1.349*** (3.81)	1.347*** (3.81)	1.358*** (3.92)	1.348*** (3.82)
Ideal age to be parent	-	0.957*** (-5.94)	-	-	-	0.956* (-2.38)
Too old to live w/ parents	-	1.004 (0.90)	-	-	-	0.999 (-0.09)
Scandinavian	-	-	0.742*** (-4.15)	-	-	0.645*** (-7.35)
Continental/Conservative	-	-	0.685*** (-4.75)	-	-	0.827 (-1.87)
Mediterranean	-	-	0.681*** (-4.09)	-	-	0.802* (-2.15)
Central/Eastern European	-	-	1	-	-	1
Anglo-Saxon	-	-	0.715*** (-3.61)	-	-	0.792* (-2.06)
Maternity leave duration	-	-	-	1.009** (2.98)	-	1.008* (2.23)
No guaranteed childcare	-	-	-	-	1	1
Guaranteed childcare	-	-	-	-	1.093* (2.11)	1.309** (3.10)
Free childcare	-	-	-	-	1.469*** (4.47)	1.152 (1.24)
Male × Ideal age	-	-	-	-	-	1
Female × Ideal age	-	-	-	-	-	0.970* (-2.39)
Male × Too old	-	-	-	-	-	1
Female × Too old	-	-	-	-	-	1.005 (0.60)
Male × Age left home	-	-	-	-	-	1
Female × Age left home	-	-	-	-	-	0.984 (-1.63)
Observations	11029	11029	11029	11029	11029	11029

**Figure 1.** Kaplan-Meier survival curve of becoming first-time parents after leaving one's parental home, by welfare regime type.**Figure 2.** Kaplan-Meier survival curve of becoming first-time parents after leaving one's parental home, by national paid maternity leave duration.

duration for two dummy variables for whether the respondents' country of residence guarantees and provides free national early childcare services. The model was similar with respect to individual-level covariates as the previous models, and when it came to the main macro-level variables of interest, revealed that both guaranteed and free childcare is positively associated with the risk of becoming a first-time parent. Compared to respondents who live in countries without a guaranteed and free place, those who have a national guaranteed and free place are at a 9% and 47% higher risk of having a first child respectively, keeping other things constant. From the Kaplan Meyer survival curves (Figure 1), we see that childcare availability and affordability appears to matter a great deal during the first few years since leaving parental home, but less so after about 15 years since the parental home.

Model 6 brings all of our previous covariates together and includes three additional interaction terms to test whether ideal age to become parent, age too old to still live with parents, and age at which respondents left home differed by genders. We found that generally, the interaction terms were not significant, apart from the slightly differential effect of ideal age to become parent on the risk of becoming a parent by genders, as mentioned above. Apart from this, Model 6 revealed that the effect of maternity leave duration remained more-or-less stable when adjusting for additional macro-level covariates. The previously highly significant and large effect of free childcare on the risk of becoming a first-time parent is now smaller and no-longer significant, while the effect of guaranteed childcare is now larger and more significant, pointing to a higher risk of becoming a first-time parent of 31% for those

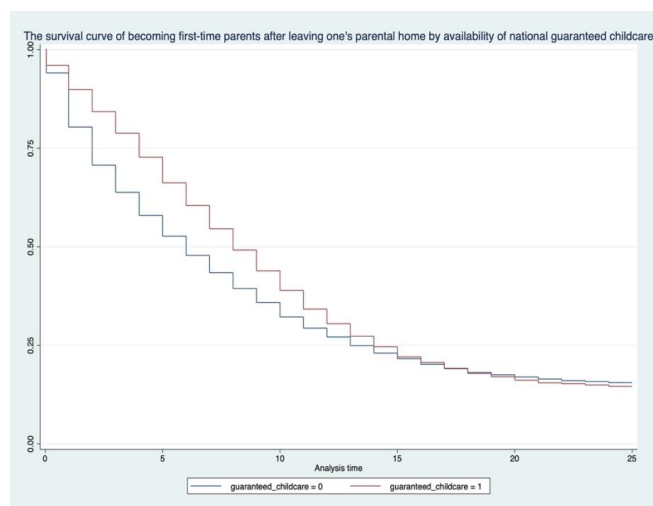


Figure 3. Kaplan-Meier survival curve of becoming first-time parents after leaving one's parental home, by availability of national guaranteed childcare.

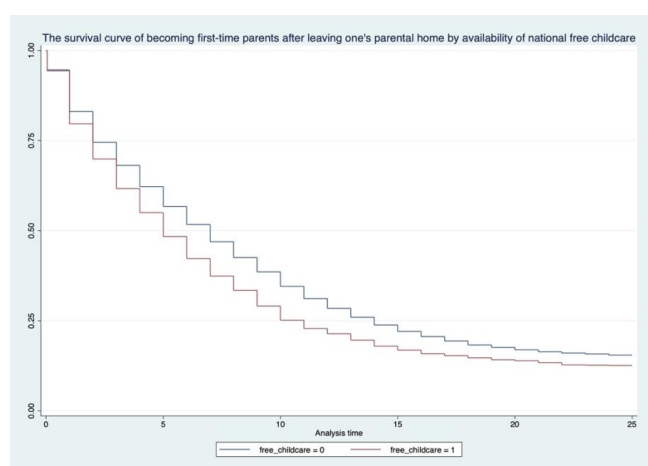


Figure 4. Kaplan-Meier survival curve of becoming first-time parents after leaving one's parental home, by availability of national free childcare.

living in countries with guaranteed childcare, compared to those without. Moreover, adjusting for maternity leave duration and childcare significantly negatively affected the significance of the estimates for welfare state regimes, pointing to the possibility that part of the variation previously explained by welfare regime clustering was now explained by included specific policy measures with higher substantive value to both researchers and policymakers.

4 Challenges and Limitations

While our analysis benefited from a large sample size encompassing a large number of countries and time periods, which allowed us to use comparative social policy and event history methodologies, the study doesn't come without a fair share of challenges.

Firstly, the construction of our duration variable relied on a survey question question 'In what year, if ever, did you first

leave your parent(s) for 2 months or more to start living separately from them?', which may be either very far away in the past and thus subject to potential recall bias, or the duration could be overestimated, because individuals may have, for example, left for college at a younger age and then returned to live with their parents at a later date.

Secondly, some of the variables used in our analysis are limited in their ability to capture the dynamic and ever-changing social context. For example, gender is defined in binary terms, leaving out a potentially significant and informative proportion of individuals who do not self-identify in binary terms without a voice. Additionally, questions related to race or immigration status were not considered in the dataset and were therefore omitted from the regression analysis.

Our most important challenge, however, pertains to the static nature of some of our variables, fixed at the value measured at the time of the survey, in 2018. For example, we used household income and occupational status at the time of interview, as opposed to a more appropriate time-varying covariate method, for which we did not have available data. However, based on extensive empirical evidence of the strong association between childhood and adult socioeconomic status, we hope that using its static proxies, income and occupational class, will allow us to approximate the life course socioeconomic effect (Nettle and Bateson 2017). Similarly, due to current constraint on resources and time, we have only used a static measure of our macro-policy variables, fully acknowledging that this relies on a faulty assumption of static family policies in Europe across multiple decades. Future research should improve upon the methodology of this paper by constructing a time-varying covariate for maternity duration, guaranteed childcare, and free childcare policies.

5 Conclusions

Using event history analysis methods, our paper attempted to shed light on the risk-factors for becoming a first-time parent after leaving parental home in Europe. We found that individual-level sociodemographic characteristics of respondents consistently affect the risk of becoming a first-time parent, whereby the risk is lower for males, individuals who have never been in a formal relationship, with more years of education, higher mother's education and higher occupational class, compared to females, those who declared a formal relationship in the past, with lower personal and mother's educational attainment and lower occupational class, respectively. Additionally, we found that macro-level indicators of family-friendly policies matter strongly for the respondents' decision to become parents and more generous benefits are associated with a higher probability of having a first child following a move away from parental home. Our findings highlight an important additional primary implication of the UNICEF's family policy recommendations beyond their effect on the wellbeing and attainment of children. Namely, it is important to realize the potential of family policies to affect the decision of individuals to become parents at a given time, in the first place and the need to examine fertility decisions and family transitions through the lens of access to affordable family-friendly policies.

Acknowledgements

I acknowledge the help of Dr. Celine Le Bourdais, who have provided me with valuable feedback on the proposal draft and presentation of this paper.

References

- Aspalter, C. (2020). *Ideal types in comparative social policy*. Routledge.
- Buck, N., & Scott, J. (1993). She's leaving home: But why? An analysis of young people leaving the parental home. *Journal of Marriage and Family*, 55(4), 863–874.
- Chzhen, Y., Gromada, A., & Rees, G. (2019). *Are the world's richest countries family friendly?* OECD Office of Research.
- Dewilde, C., & Lancee, B. (2013). Income inequality and access to housing in Europe. *European Sociological Review*, 29(6), 1189–1200.
- Eikemo, T. A., Bambra, C., Judge, K., & Ringdal, K. (2008). Welfare state regimes and differences in self-perceived health in Europe: A multilevel analysis. *Social Science & Medicine*, 66(11), 2281–2295.
- Esping-Andersen, G. (1990). *The three worlds of welfare capitalism*. Polity Press.
- Esping-Andersen, G. (Ed.). (2007). *Family formation and family dilemmas in contemporary Europe*. Fundacion BBVA.
- European Social Survey. (2018a). *Sampling / European Social Survey (ESS)*.
- European Social Survey. (2018b). *Survey specification / European Social Survey (ESS)*.
- Ferrera, M. (1996). The 'Southern model' of welfare in social Europe. *Journal of European Social Policy*, 6(1), 17–37.
- Frejka, T., & Sardon, J.-P. (2006). First birth trends in developed countries: Persisting parenthood postponement. *Demographic Research*, 15, 147–180.
- Haggard, S., & Kaufman, R. R. (2009). The Eastern European welfare state in comparative perspective. In *Post-communist welfare pathways* (pp. 217–236). Palgrave Macmillan.
- Heaton, T. B., & Call, V. R. A. (1995). Modeling family dynamics with event history techniques. *Journal of Marriage and Family*, 57(4), 1078–1090.
- Martin, C. (2015). Southern welfare states: Configuration of the welfare balance between state and the family. In *Post-communist welfare pathways*. Palgrave Macmillan.
- Motiejūnaitė-Schulmeister, A., Balcon, M.-P., & De Coster, I. (2019). *Eurydice brief: Key data on early childhood education and care in Europe*. European Commission.
- Mulder, C. H., Clark, W. A. V., & Wagner, M. (2002). A comparative analysis of leaving home in the United States, the Netherlands and West Germany. *Demographic Research*, 7, 565–592.
- Mulder, C. H., & Hooimeijer, P. (2002). Leaving home in the Netherlands: Timing and first housing. *Journal of Housing and the Built Environment*, 17(3), 237–268.
- Nettle, D., & Bateson, M. (2017). Childhood and adult socioeconomic position interact to predict health in mid life in a cohort of British women. *PeerJ*, 5, e3528.
- OECD. (2020). *Parental leave systems*. PF2.1.
- Studer, M., Liefbroer, A. C., & Mooyaart, J. E. (2018). Understanding trends in family formation trajectories: An application of Competing Trajectories Analysis (CTA). *Advances in Life Course Research*, 36, 1–12.
- Van Hook, J., & Altman, C. E. (2013). Using discrete-time event history fertility models to simulate total fertility rates and other fertility measures. *Population Research and Policy Review*, 32(4), 585–610.

Appendix A: Country classification by welfare regime type and the state-market-family division of roles

Welfare Regime	Countries	State Role	Market Role	Family Role
Anglo-Saxon	Great Britain, Ireland	Marginal	Central	Marginal
Continental/Conservative	Austria, Switzerland, Germany, Belgium, France, Luxembourg	Subsidiary	Marginal	Central
Scandinavian	Denmark, Sweden, Finland, Iceland, Norway, Netherlands	Central	Marginal	Marginal
Mediterranean	Cyprus, Italy, Spain, Turkey, Portugal, Greece	Marginal	Subsidiary	Central
Central/Eastern European	Slovakia, Bulgaria, Czech Republic, Estonia, Poland, Croatia, Hungary, Lithuania, Slovenia, Romania	Central	Marginal	Subsidiary

Table generated by the author of this paper. Sources: Esping-Andersen 1990, Martin 2015, Beblavy 2008, Haggard and Kaufman 2009.

Appendix B: Summary Statistics by Welfare Regimes

Full stratified summary statistics (mean, SD, N, range) by welfare regime for marital status, age left home, household income, birth year, gender, years of education, mother's education, occupational class, maternity duration, ideal age for first-time parenthood, and age too old to live with parents are available from the author upon request, and were reported in full in the original submitted manuscript.

Appendix C: Summary Statistics by Birth Cohorts

Full stratified summary statistics by birth cohort (1929–1949, 1950–1969, 1970–1989, 1990–2009) for the same set of variables are available from the author upon request, and were reported in full in the original submitted manuscript.

Appendix D: Additional Kaplan-Meier survival curve

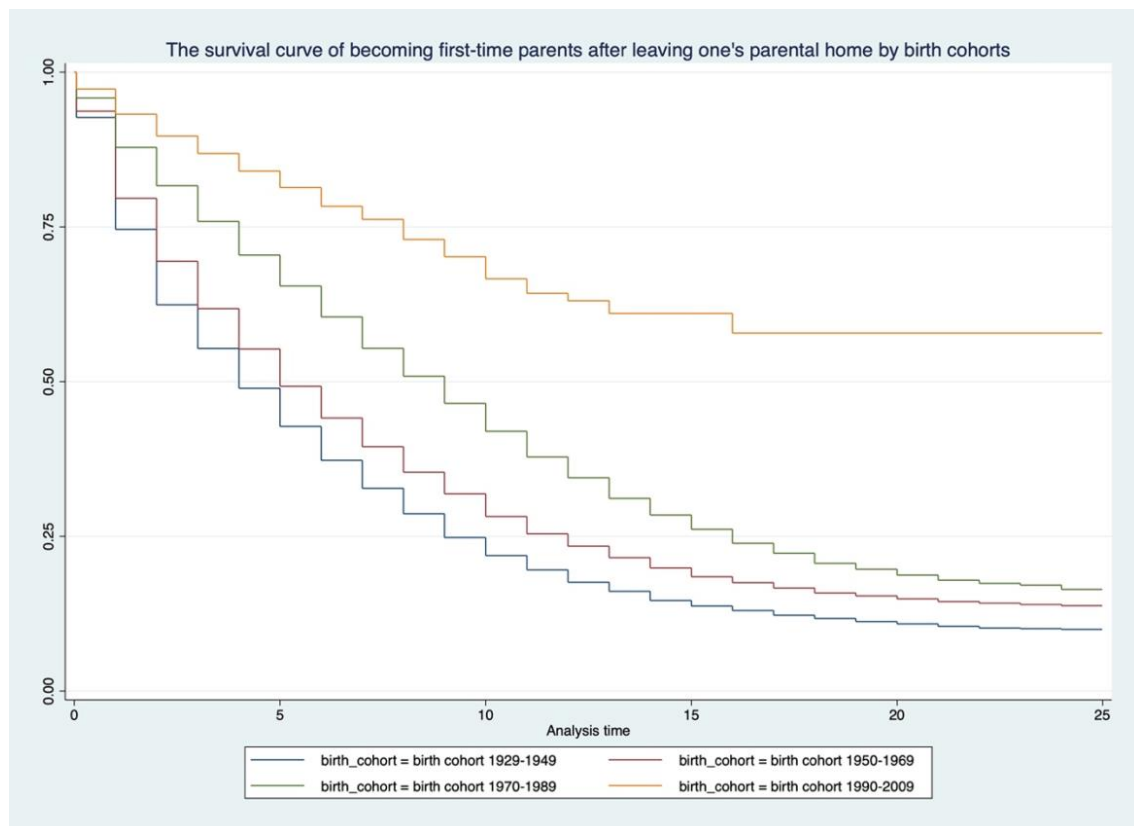


Figure 5. Kaplan-Meier survival curve of becoming first-time parents after leaving one's parental home, by birth cohort.